

THE JOY OF CLIMBING

As a conclusion to this book, we would like to highlight the importance of having a good training environment. There are many factors that come into play here. Besides the purely practical elements, such as good climbing walls and outdoor crags, inclusion, participation and mastery are key words for a good training environment. By giving everyone space, participating with and praising others, you can be part of a community that creates and promotes good climbers. By helping others, there is a great chance that the training community will also help you.

For example, there is the importance of the dialogue we have with each other to improve our technique. Whenever we are bouldering indoors, we constantly discuss beta with each other. We discuss whether a move should be done statically or dynamically, and we discuss foot placements and grip positions. ‘Are you pinching? I’m crimping’, ‘Use more speed’, and ‘Relax and lower your body’ are all examples of statements made countless times in the course of a session. The same dialogue is also important when it comes to the mental and physical elements. By supporting each other and rationally going through what happens when the rope takes on a lead fall, we will more easily build the confidence needed to dare to fall. When we are doing pull-ups, it also helps to have a friend supporting you and pushing you. By participating in and being inclusive in such a dialogue, you will foster an environment where everyone helps each other out – and it is wonderfully motivating to be a part of such an environment.



Don't get demotivated by strong climbers! If you're lucky enough to climb alongside stronger climbers, use them as a source of motivation – something to strive for. Chances are you'll push harder and get better because you are climbing with them.

Focus on mastery and not just on results. This applies to both yourself and your own achievements, and when rewarding those around you. A performance-oriented environment where competition and individual performance are rewarded tends to foster athletes who are afraid of failing and do not dare to take on new challenges. Such athletes often end up quitting and this builds a negative and not very inclusive training environment. By appreciating your own and others' progress instead of just achievements, you will help create an environment that you and others will thrive in. Appreciate progressing further than before on your project, or being a little closer to managing five pull-ups, or finally managing to hang from the edges on the Beastmaker. Let others know that you are happy with your own progression, and let others know when they progress.

The intention isn't that you should not be performance-oriented at all. To a certain extent, focusing on achievement is important for becoming a better climber, but it must not overshadow the joy of getting better and getting a sense of achievement. External factors, such as winning a competition or sending a difficult route, are a nice bonus, but it's often those who enjoy the little things, like getting better at pinching or sticking a move they haven't managed before, who will remain climbers for their entire lives. They are also often the ones who become the best, and definitely the ones who have the most fun.

Some people have the ability to lift their own and others' achievements. You can be this person by being inclusive, participatory, mastery-oriented and positive. In order to illustrate a good training environment, we want to highlight some periods and communities from our own climbing lives that have contributed to our development and defined our identity as climbers.





THE BEGINNING

We were 13 to 14 years old when we started climbing. The community was much smaller than it is today, and the natural consequence of this was that anyone who started got included in this community from day one. We started at Skøyen Climbing Centre, and the community there, spearheaded by Jacob Normann, took us under its wings. We got to go climbing at Damtjern and Sørkedalen and we felt like an important part of the community. Being included in this way made us super stoked, and we climbed almost every day. We learned technique by watching others and by getting feedback – everyone helped everyone. We felt a sense of achievement, and it was an incredibly fun time with a lot of progression and experiences.

This early period created an important part of our identity as climbers, and without this good start, it's not certain that we would still be climbing. Our history is an indicator of how important it is to create a good and including environment. Especially for the young! After all, we want them to have as much fun with this amazing sport as we have.

TO HELL AND BACK

A trip that stands out is when we went to Hell when we were 15 or 16 years old. We caught a flight to Værnes, did some grocery shopping at the store next to the airport and pushed a shopping trolley with our food and luggage to the parking area near the crag where we camped below the cliff for a week. The climbers who passed by liked what we did, gave us lots of tips on routes to try and we got to know a lot of new people. After that, we competed at a Norwegian lead cup event at Dragvoll in Trondheim. Stian climbed wearing a Superman outfit, and afterwards we partied with a crazy bunch of climbers. Memories for life!



COMPETITIONS

As we got older we started climbing in a lot of competitions, both national and international. For a long time we had a coach, Miguel De Freitas, who came with us to competitions all around the world. Miguel wasn't a strong climber himself, and he didn't teach us a lot about technique, but he introduced us to several invaluable methods

for endurance and strength training. He was there with us and created a sense of security and belonging, and he was good at pointing out achievements instead of results.

We were never really that good at climbing competitions, but that is probably because we longed for the lifestyle of outdoor climbers. Competitions were the path young climbers were expected to take, but were perhaps not the right path for us. Our identity was anchored in outdoor climbing from day one. And we missed the outdoor climbing community we had been a part of since the beginning. The moral of this is perhaps you will only perform at your very best and feel 100 per cent comfortable if you know you're in the right place. The minute we started focusing on outdoor climbing again, we trained harder than ever and we never doubted which path we had taken.



PHOTO: PAUL ROBINSON

Mirage (Font 8a+), Driehoek, South Africa.

NOW

Jumping back to the present, we're now adults and have partially made climbing our jobs, and we still climb as much as possible every week. We sometimes climb in competitions, but we regard ourselves as outdoor climbers – who spend a frightening amount of time training indoors ... ! Since the time being filled with competitions, we

have mostly been climbing non-stop. There have of course been periods with less climbing because of injuries or other things in our lives, but the community has always been an important part of our identity. What still motivates us most is spending a day at the crag with good friends who all share the same passion. A new Jacob or Miguel has entered the scene on multiple occasions, shaping the path forwards and making a positive contribution to the creation of a good environment. Perhaps we have been these guys ourselves?

The last person who's had a big impact on our own training environment is Kenneth Elvegård. Kenneth started climbing in 2011 and got good fast. He arrived in an Oslo scene which many felt was overly focused on performance and not very inclusive. Coming from other sports, Kenneth wasn't accustomed to this, and by being stoked, positive and inclusive he managed to influence and change the Oslo environment he now was a part of into the good community it is today. This might sound like something of an oversimplification, but sometimes this is just the way it is. We have discussed this at length and have come to the conclusion that sometimes all it takes to turn an environment on its head is just one stoked person. Mood and attitude can be incredibly contagious – they can disarm failure and get everyone to perform at their best. They can remove negative competition, and the collective will then drive all climbers forwards.



PHOTO: BJØRN HELGE RØNNING

Good vibes during the wedding of Martin Mobråten and Maria Davies Sandbu.

WHAT HAVE WE LEARNT?

As we have seen, there are several things that can make a good training environment. Sometimes we're lucky enough to see one individual set a new course, but we'll go far if we're all inclusive, participate, promote mastery and are stoked.

There is an obvious connection between climbers who find joy in climbing and keep it as an important part of their lives for a long time, and those who value mastery and participate in an inclusive environment with shared values. We write this as a prayer to all climbers, coaches and parents: don't give too much thought to results, either in competitions or the climbing of a hard grade. Build an environment and give positive feedback when friends, children or athletes learn something new, do their first pull-up, or dare to fall on lead for the first time.

THE TEN COMMANDMENTS OF CLIMBING

1. You have to climb a lot to be a good climber.
2. Vary your climbing between different styles and angles – both indoors and outdoors.
3. Train technique before physical training.
4. Learn to use your feet – they will be your best friends on the wall.
5. Rid yourself of the fear of falling.
6. Train finger strength.
7. Find your strengths and weaknesses, set yourself targets and adapt your training thereafter.
8. What and how you think is crucial to your success – become just as strong mentally as you are physically and technically.
9. Create – or become part of – a supportive environment that challenges you.
10. Preserve the joy – climbing is all fun and games.



PHOTO: TERJE AAMOT

EPILOGUE

The more we learned, read and taught over the years, the more apparent it became that there was a need for a comprehensive and accessible book on the subject of training for climbing. The turning point, when the process transformed from thoughts and ideas into an actual project, came during a trip to Vingsand in January 2017, when Martin simply declared ‘We’re just going to do it’. The road from that day in January to now with this book in your hands has been long and extremely educational. Because – where do you start when you want to write a book?

First and foremost, we wanted to write a book that anybody who wants to read about climbing can understand, with helpful hints and tools that are easy to use, even for those with limited climbing experience. This has been a challenging process because writing simply and clearly about complicated and comprehensive topics is difficult and a bit intimidating. As Albert Einstein put it: Make it as simple as possible, but not simpler. Every chapter in this book could have been a book in and of itself. We therefore had to leave out a few elements, and simplify others, but we hope that we have still managed to capture the essence of the topics we deemed most important.

Secondly, we wished to write a book that was both educational and inspirational. We have put great emphasis on using good photography, and we want to thank all of the photographers who were generous enough to help us create a visually appealing book: Bård Lie Henriksen for the fantastic job of taking, sorting and editing all of the 4,000 photos that were the basis for the instructional pictures. Nathan Welton, Terje Aamodt, Alex Manelis, Chris Burkhard, Henning Wang, Bjørn Helge Rønning, Andy Wickstrom, Paul Robinson, Endre Vik, Ragnhild Tronvoll, Bjørnar Smestad, Bjørn Sæt看an, Katrine Vandet Salling, Dag Hagen, Audun Bratrud, Mike Mullins, Ryan Waters, Jon Tore Modell and Volker Schöffl for all the other photos, which we hope inspired you as much as they did us.

To finish a book with the somewhat ambitious title of The Climbing Bible is not a two-man project. Our editor, Maria Stangeland, has done a fantastic job of directing two climbing nerds in the right direction with regards to content, phrasing and substance. We have also received invaluable help with proofreading and professional assessments from a number of people in different areas, and would like to thank:

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Big thanks also to Jo Nesbø for writing the foreword to this book. This was a great honour!

To illustrate certain topics and inspire you, we added the segments we called ‘A Climber’s Story’. In them, Magnus Midtbø, Rannveig Aamodt and Cecilie Skog gave us fantastic depictions from their own lives as climbers. In addition, Kenneth Elvegård and Joakim Louis Sæther have both written valuable contributions on route setting and jamming, respectively. Eva López has elaborated on her thoughts on finger training, Tom Randall has given us more information about his approach to training structure, and Tina Johnsen Hafsaas, Thilo Schrøter and Hannah Midtbø have shared their pro tips with us. Thank you for taking the time to share your stories and knowledge with us.

Thanks also to Jon Tore Modell and Elisabet Skårberg at Spanish Inq for a fantastic collaboration with design and layout, and for believing in us and helping us to create exactly the book we had imagined.

More generally, we would also like to thank the wonderful climbing community that we are part of. For 20 years now we have been inspired by, challenged by and had long discussions with a number of people about this and that. All these experiences have helped in shaping this book. We are very pleased to see that climbing is a growing sport and that more and more people are experiencing the same magic that hooked us so many years ago. The future of climbing in Norway and internationally is brighter than ever.

CLIMBING GRADES – BOULDERING

V GRADE (V)	FONTAINEBLEAU GRADE
VB	3
V0-	3+
V0	4
V0+	4+
V1	5
V2	5+
V3	6a / 6a+
V4	6b / 6b+
V5	6c / 6c+
V6	7a
V7	7a+
V8	7b / 7b+
V9	7c
V10	7c+
V11	8a
V12	8a+
V13	8b
V14	8b+
V15	8c
V16	8c+
V17	9a

CLIMBING GRADES – ROUTES

USA	FRANCE	UIAA	NORWAY	BRITISH	
5.0				Mod VD HVD S HS	3c 4a 4b
5.1	1	I	1		
5.2	2	II	2		
5.3	3	III	3		
5.4	4a	IV	4		
5.5	4b	IV+	5-		4b
5.6	4c	V	5	VS	4c
5.7	5a	V+		HVS	5a
5.8	5b	VI-	5+		
5.9	5c	VI	6-	E1	5b
5.10a	6a	VI+			
5.10b	6a+	VII-	6	E2	5c
5.10c	6b	VII	6+		
5.10d	6b+	VII+	7-	E3	
5.11a	6c 6c+		7	E4	6a
5.11b		VIII-			
5.11c			7+		
5.11d	7a	VIII		E5	6b
5.12a	7a+	VIII+	8-		
5.12b	7b				
5.12c	7b+	IX-	8	E6	6c
5.12d	7c	IX			
5.13a	7c+	IX+	8+		
5.13b	8a		9-	E7	
5.13c	8a+	X-	9-/9		
5.13d	8b	X	9	E8	7a
5.14a	8b+	X+	9/9+		
5.14b	8c		9+	E9+	7b
5.14c	8c+	XI-	9+/10-		

5.14d	9a	XI	10-	
5.15a	9a+	XI+		
5.15b	9b	XI+/XII-		
5.15c	9b+	XII-		

GLOSSARY

Approach: The walk-in to the climbing area.

Anchor: Usually two bolts connected with a chain which marks the end of a route.

Arête: Outside corner/edge of two walls, like the outside corner of a building.

Back-clip: Clipping a quickdraw incorrectly, so that the rope is routed from the outside and in, rather than from the inside and out. Creates the risk of the rope unclipping if the climber falls.

Barn door: When one of your legs swings out from the wall like a crooked door because you're in an off-balance position as you move to the next hold.

Belay device: This helps you hold/ control the rope when belaying your partner, and your partner either falls or asks to be lowered to the ground.

Belayer: The person belaying (holding the ropes of) the climber.

Beta: Usually describes information about moves or sequences, but can also refer to travel, accommodation, weather, and so on.

Bicycling: A version of the toe hook where you stand on the front of a foothold with one foot while toe hooking on the back with the other.

Boulder problem: A defined route up a boulder or short wall where no rope is needed. Indoors it will usually start and stop on marked holds, but outdoors it starts at the holds you can reach from the ground and finishes by topping out the rock (unless otherwise stated in the guidebook, for example sit-starts, problems which only go part way up a wall, or problems with rules).

Brush stick: A long stick with a brush at one end which allows you to brush holds beyond your normal reach.

Brush: Used to remove chalk and dirt from climbing holds. Toothbrushes have long been used for this, but in recent years a number of different brushes have been specially made for climbing. (Brushes with real bristles from boar or the like are the best.)

Brushing: cleaning chalk and dirt from the holds.

Bumping: Grabbing a hold, and then immediately using it to generate speed to grab the next hold with the same hand.

Campus board: A training device made up of wooden rungs on an overhanging plywood wall.

Chalk bag: A bag of chalk attached with a belt around your waist. There are also bigger chalk buckets placed on the ground for bouldering when you don't need to carry a bag during the climb.

Chalk: Chalk is used to prevent the sweat in your hands from ruining the friction. The chalk we use isn't actually chalk, but magnesium carbonate. You can buy regular chalk, liquid chalk or chalk balls.

Chalking up: To apply chalk to your fingers and hands or the holds before or during a climb.

Chipping: Making new or improving existing holds by using tools like a drill, chisel, hammer, glue or cement. Considered unethical and should be avoided.

Cleaning: To ready a route or boulder for climbing. This involves among other things the removal of moss and loose holds by use of a stiff brush or similar tools. Cleaning a route can also mean removing the quickdraws after climbing it.

Clip stick/beta stick/cheat stick: Specialised tool-on-a-stick that allows you to clip a quickdraw higher than you can normally reach. Often used to pre-clip the first bolt on sport climbs so as to avoid a ground fall.

Clipping: When you attach the quickdraw to the bolt hanger, or the rope through the quickdraw when lead climbing.

Clutch: Immediately moving your first hand to the next hold as your second hand lands the first hold because the first hold isn't good enough on its own.

Compression: Squeezing your hands together on two opposing holds. A prow with slopers in bouldering is known as a compression problem or fridge climbing (because it's like lifting a fridge).

Corner: AKA dihedral. The inside corner of two walls, like the inside corner of a room. Opposite of arête.

Crag: The cliff or boulders where the climbing is.

Crash pad/bouldering mat: A portable mattress filled with layers of foam. Used to soften the landing when bouldering. Usually carried like a backpack on the approach.

Crimp: Adding your thumb over your index finger to lock the grip down.

Crossing: When you cross your right hand to the left, past your left hand, or vice versa, you're crossing over. You can cross over with your foot or your hand, above or below, behind or in front, depending on the holds and your body position.

Crux: The hardest move or sequence of a route or boulder problem.

Dab: When you accidentally touch the ground, the crash pad, a spotter or anything else nearby that isn't part of the climb. Dabbing invalidates an ascent.

Deadhang: Finger strength training method where you hang from the edges of a fingerboard or similar.

Deadpoint: Stopping your body at the apex of a dynamic movement. Refers to the point in the trajectory where the motion 'dies', i.e. transitions from going up to going down, and you are practically weightless.

Dogging: Stopping and hanging on the rope at each bolt when lead climbing so you can figure out the beta.

Dry-fire: When you slip off a hold because your fingers are either too dry or too cold for there to be enough friction. Often happens without notice; can be painful and wears skin.

Dynamic: Doing a move dynamically involves using the whole or parts of the body to create momentum and swing towards the next hold, instead of locking off and moving just the arm.

Dyno: A dynamic move where you jump to and catch a hold or holds with one or two hands while both feet cut loose.

Edge: A hold so shallow that only the outermost pads of your fingers can fit on it.

Edging: Using the side edges of your climbing shoe to stand on small edges. Stiffer shoes are better.

Elvis leg: Or sewing machine leg. When your leg starts vibrating up and down like the needle of a sewing machine, often because you are tired or stressed during a climb. Makes standing on smaller footholds difficult.

Fingerboard: A board with a selection of edges and holds for training arm and finger strength.

First ascent: When you are the first person to successfully climb a route or boulder problem, you have done the first ascent, often abbreviated to FA. It is often the person who has cleaned and/or bolted a line who does the first ascent, but not always. This person is known as the creator, and until the route or boulder problem has been climbed, it is called a project.

Flash: Climbing a route or boulder problem first go, which you've never tried before but for which you have beta on how to solve the different moves and sections.

Flick: Moving your lower hand just as your upper hand lands the next hold.

Friction: The force acting between two surfaces that are touching, such as your hand or foot on a climbing hold: good or bad friction refers to how it feels when you are holding or standing on a hold. The friction is dependent on the texture of the hold, the contact surface between the hold and your skin/boot rubber.

Frogging: Squeezing both feet in towards one other with your knees pointing outwards. Often used on arêtes, tufas or opposing footholds placed next to one other.

Fronting: Standing with the inside edge of the foot on the foothold so that the inside of the leg is facing the wall. The knee is pointing out to the side and away from the body.

Gaston: See shoulder press.

Go: Synonymous with attempt. Having a go means an attempt from the start with the intent of success.

Grade: The grade of a route or boulder problem indicates the difficulty or how challenging it is to climb. Different countries have different grading systems for routes and bouldering.

Gym: Indoor climbing wall/centre.

Harness: Usually consists of a waist belt (with gear loops), a belay loop and leg loops. The harness is what you attach your rope and/or belay device to when sport or trad climbing.

Heel hook: Hooking your heel on or behind a hold or edge.

Height-dependent: AKA morpho. Usually refers to routes or boulder problems where being tall is an advantage because of the distance between the holds. But remember, being short can also be an advantage. For example, it's not easy getting your feet high underneath holds if you're tall. Morpho can also mean it's easier for shorter climbers, but this is rarely used.

High point: The furthest up a route or boulder problem you have managed to get before a successful ascent.

Highball: A tall boulder problem.

Incut: Edges that have a positive angle in relation to the angle of the wall.

Indoor climbing: Climbing inside at a climbing wall/centre.

Jamming: Locking your fingers, hands, feet or other body parts in cracks in the rock.

Jug: A really good hold.

Kneebar: If the distance between a foothold and a large hold or opposing surface is roughly the same length as your lower leg, with your foot on the foothold you can squeeze your knee against the large hold or opposing surface. This will in many cases allow you to significantly reduce the load on your arms and fingers, and sometimes even allow you to completely let go with your hands. Recent years have seen the emergence of rubberised knee pads, which increase friction and make it possible to use smaller and poorer kneebars.

Layback: Leaning away from the holds to increase grip so you can hang on. A technique often used on cracks or arêtes where the grip surface is vertical.

Lead: Climbing on lead means clipping the rope into the quickdraws on the way up. In competitions, all quickdraws must be clipped, but outdoors only those that are necessary to avoid long or dangerous falls are used. For an ascent to be valid the route must be climbed on lead without loading the rope for a rest.

Line: Synonym for route or boulder problem, describing where to climb up the wall. We often say this or that line stands out or looks good, and this is related to which part of the rock face looks most inviting to climb.

Lock-off: When you stop all motion in your body, essentially locking it off, and move one hand to the next hold. You can train this skill by doing lock-off exercises.

Lowering: When your belayer lowers you to the ground.

Mantelshelf/Manteling: Pushing yourself up with one or two hands, and ‘standing’ on them with your elbows straight. Mostly used for topping out outdoors.

Matching: Placing both hands or feet on the same hold.

Moonboard: Training wall developed by Ben Moon with a specific layout of holds and problems.

On-sight: When you try a route for the first time without any prior information other than the grade.

Open hand: All grip positions where the angle of the middle finger joint is greater than 90 degrees.

Overhang: Walls where the angle is steeper than vertical.

Pendulum: When your body swings into the wall when you take a fall on lead as the rope takes, or when your body swings as you dyno to a hold and your feet cut loose on steep terrain – holding a swing like this requires strength in the fingers and upper body.

Pinching: Squeezing a hold between your thumb and fingers. A pinch is a hold that invites such a grip.

Pocket: A hole with room for one or more fingers.

Pogo: Using one leg as a pendulum to create speed and momentum to execute a dynamic move.

Power spot: Pushing a climber up and into position to try a move on a boulder problem.

Pre-clip: When falling at the start of a route is deemed too dangerous, you can pre-clip one or more quickdraws, so that the start is climbed on a top rope. Whether to pre-clip and, if so, how many quickdraws, is an ethical question, as this can change the grade and character of a route. Consensus regarding pre-clipping of the route in question usually comes down to the local norms.

Project: A route or a boulder problem you're trying. **Warning:** a project can easily morph into an obsession. Outdoor routes or boulder problems that have been cleaned and bolted but have not yet had a first ascent are also known as projects. If the creator wants to climb it first, the project is closed, but if anyone can have a go at the first ascent, the project is open.

Pulling through: When you don't stop to match a hold but continue the upward motion with your other hand.

Pump: the feeling of getting tired in your forearms.

Quickdraw: Two karabiners connected together with a short sling. One karabiner is clipped to the protection or bolt, and the rope is clipped through the other karabiner.

Rack: Common term for all the gear you carry with you on a climb; usually safety gear like belay device, quickdraws, nuts and cams, slings, and so on.

Rappelling: Lowering yourself down from the top of a route.

Reading a route/boulder problem: Looking at the route and making a plan for which holds to use with which hand, and where to place the feet. Mainly used before an on-sight attempt.

Redpoint: climbing a route on your second attempt or later is known as a redpoint. The expression comes from Frankenjura, Germany, where climbing legend Kurt Albert painted a little red dot – a 'rotpunkt' – at the start of any route he had done without holding or standing on anything other than the rock.

Rest: A good hold or a good position on the wall where you can stop, rest and shake some of the pump out of your forearms. Here you can psyche yourself up or calm yourself down, depending on what comes next.

Rocking over: Sitting down on one leg by fully bending the knee. A useful technique for vertical climbing as it significantly lessens the load on the fingers and arms.

Roof: A horizontal overhang.

Rope drag: The longer the route, the more contact points the rope has against quickdraws and protrusions on the wall, and this all creates more resistance – or rope drag. Rope drag is also a result of the weight of the rope and the friction between the rope and the quickdraws and the wall.

Rope gun: Describes the strongest climber of the crew – the person leading and fixing top ropes for everyone else.

Rope team: Consists of a belayer and a climber.

Rope: Climbing ropes are made up of an elastic core and a protective sheath, and come in different lengths and thicknesses depending on the application.

Route: A route is a defined way up the climbing wall, either indoors or outdoors. It can be a bolted line or a trad line, and it starts from the holds you can reach from the ground and finishes at an anchor or by topping out.

Run-out: If the bolts are few and far between. Can lead to big falls.

Safety check: Climber and belayer check each other's gear before climbing. Always do a pre-climb safety check!

Sandbag: A route that is harder than the grade suggests. Grades differ depending on the route, the wall, the crag, or the country you're in. Some routes will feel 'soft' (easy), while others will feel 'hard'. Some destinations have been referred to as having 'holiday grades', while others are not recommended if you want to climb a higher grade. Remember that grades are only opinions and these opinions will differ depending on a climber's height/reach, finger size and strengths/weaknesses.

Sending: When you successfully climb a route or boulder problem, you've sent it.

Shoe: Short for climbing shoe. AKA rock boot. Shoes specially made for climbing, with distinctive blends of rubber compounds for maximum friction and/or durability. There are many different climbing shoes for different purposes.

Shoulder press: A vertical hold where the grip surface is facing your body can be used as a shoulder press. Also known as gaston.

Sidepull: A vertical hold where the grip surface is facing away from your body.

Sit-start: Shorter boulder problems often have a sit-start. Your bottom should be the last part of your body to leave the ground.

Skipping: When you miss clipping a quickdraw.

Slab: Walls where the angle is shallower than vertical.

Sloper: A rounded hold with a negative angle in relation to the angle of the wall.

Smearing: Standing on a rounded foothold without does not have any distinct edge. Imagine smearing the rubber on to the hold or wall, like spreading peanut butter on a sandwich. Softer shoes are better suited to smearing.

Speed climbing: A category within competition climbing, where the goal is to climb a standardised route as fast as possible on a top rope.

Spotting: Standing at the ready to brace/support a bouldering climber in the event of a fall.

Static: Doing a move statically means moving slowly and in control, without building momentum in advance.

Stemming: Pushing your hands and feet on to opposing walls in a corner/ dihedral or other similar features.

System board: A training wall with a specific layout of holds.

Takes skin: A route or boulder takes skin if there are one or more sharp holds that wear skin fast. Such routes or boulders should be climbed on cold days with good friction, and you need to be careful to limit the number of attempts.

Tick/tick mark: A tick mark is a thin stripe of chalk, indicating where to grip or stand on a hold. Ticking is the act of making this stripe. Ticks should always be removed before you leave a route or boulder.

Ticklist: A list of routes and boulders you have climbed or would like to climb.

Toe hook: Hooking the upper side of your foot behind a hold or an edge.

Top rope: When you are belayed from the top of the route, either via an anchor at the top or a belayer above you, you are climbing on a top rope. It's common to start top roping before learning to lead climb.

Topping out: Synonymous with success. Usually done literally in bouldering, where you finish by climbing on to the top of the boulder.

Tufa: A tufa is a stalactite. They can often be observed as long or short vertical lines on the wall, and they are formed when carbonate minerals precipitate out of ambient temperature water running down the wall. Usually found on softer rock types like limestone.

Twisting in: Standing with the outside edge of your shoe and twisting your leg so that the outside of your hip and leg is facing the wall. Your knee is then pointing in towards the centre of your body.

Undercling: A hold where the grip surface is facing down.

Volume: Larger holds placed on indoor walls to make three-dimensional changes to the wall. Come in all shapes and sizes. Can be used both as handholds and footholds, and smaller holds can often be placed on them.

Whipper: A long fall when lead climbing.

READ MORE

There is more knowledge to be gained about both training and climbing, so we want to encourage anyone who's feeling inspired to read and learn more about the different topics to continue with the books we have listed below for each chapter. Because, as you know, a foolish man thinks he knows everything. A wise man knows he doesn't.

PHYSICAL TRAINING:

Anderson, M. and Anderson, M., The Rock Climber's Training Manual. A Guide to Continuous Improvement, (Fixed Pin Publishing, 2014).

Hörst, E.J., Training for Climbing. The Definitive Guide to Improving your Performance, (Falcon Guide, 3rd edn, 2016).

MacLeod, D., 9 out of 10 Climbers Make the Same Mistakes, (Rare Breed Productions, 2010).

TECHNIQUE:

Anderson, M. and Anderson, M., The Rock Climber's Training Manual. A Guide to Continuous Improvement, (Fixed Pin Publishing, 2014).

Hague, D. and Hunter, D., The Self-Coached Climber. The Guide to Movement, Training, Performance (Stackpole Books, 2006).

Hörst, E.J., Training for Climbing. The Definitive Guide to Improving your Performance, (Falcon Guide, 3rd edn, 2016).

MacLeod, D., 9 out of 10 Climbers Make the Same Mistakes, (Rare Breed Productions, 2010).

Matros, P. and Korb, L., Gimme Kraft. Effective Climbing Training, (Café Kraft GmbH, 2013).

MENTAL TRAINING:

Hörst, E.J., Training for Climbing. The Definitive Guide to Improving your Performance, (Falcon Guide, 3rd edn, 2016).

MacLeod, D., 9 out of 10 Climbers Make the Same Mistakes, (Rare Breed Productions, 2010).

Moffatt, J., Mastermind. Mental Training for Climbers, (Vertebrate Publishing, 2019).

Orlick, T., In pursuit of Excellence: How to win in sport and life through mental training, (Human Kinetics, 4th edn, 2008).

INJURIES:

Hochholzer, T. and Shoeffl, V., One move too many: How to understand the injuries and overuse syndromes of rock climbing. (Lochner-Verlag, 3rd edn, 2016).

Hörst, E.J., Training for Climbing. The Definitive Guide to Improving your Performance, (Falcon Guide, 3rd edn, 2016).

MacLeod, D., Make or Break. Don't let climbing injuries dictate your success, Rare Breed Productions. 2015.

PLANNING FOR TRAINING

Anderson, M. and Anderson, M., The Rock Climber's Training Manual. A Guide to Continuous Improvement, (Fixed Pin Publishing, 2014).

Hörst, E.J., Training for Climbing. The Definitive Guide to Improving your Performance, (Falcon Guide, 3rd edn, 2016).

For questions regarding specific citations, the authors can be contacted at post@klatrebibelen.com

Climbing isn't a sport that has seen much research, neither from a performance-enhancing nor from an injury-prevention perspective, but the knowledge database is ever increasing. We recommend the website www.ircra.rocks for updated research papers. We also want to recommend the websites <http://en-eva->

lopez.blogspot.com and www.trainingforclimbing.com, which both produce articles on training and injury prevention with quality citations.
